Library Progress International
Vol.44 No.3, Jul-Dec 2024: P.20211-20221

Print version ISSN 0970 1052
Online version ISSN 2320 317X

Original Article

Available online at www.bpasjournals.com

Neuromarketing: The Science of Consumer Behavior in Digital Advertising

[1]M. Sri Ram Kailash, [2]Dr. Gautam Donga, [3]Ch.Satya Keerthi.N.V.L, [4]Crispin J Fernandez, [5]Sruthi S

[1]Assistant Professor Department of Commerce SRM Institute of Science and Technology, College of Science and Humanities, Vadapalani Campus, No - 1, Jawaharlal Nehru Salai, Vadapalani, Chennai - 600026.
[2]Assistant Professor Udhna Citizen Commerce College and S.P.B. College of Business Administration and Smt. Diwaliben Harjibhai Gondaliya College of BCA and IT, Pin: 395004
[3]Assistant Professor Department of CSE Koneru Lakshmaiah Education Foundation , Vaddeswaram, A.P, India
[4]Research Scholar Department of Management Bharathiar University
[5]Research Scholar Department of Commerce Bharathiar University

How to cite this article: M. Sri Ram Kailash, Gautam Donga, Satya Keerthi.N.V.L,Crispin J Fernandez, Sruthi S (2024) Neuromarketing: The Science of Consumer Behavior in Digital Advertising. *Library Progress International,* 44(3), 20211-20221

Abstract

This paper aims to explore the role of neuromarketing in understanding and influencing consumer behavior within digital advertising. By examining how neurological insights can enhance advertising strategies, the study addresses the effectiveness of neuromarketing in optimizing engagement, recall, and decision-making processes in digital marketing contexts.

The paper is grounded in consumer behavior theories and cognitive neuroscience, focusing on how brain activity and emotional responses can be leveraged to interpret consumer preferences. The theoretical approach integrates elements from behavioral psychology, marketing theory, and neuroscience to outline how consumers process digital advertising stimuli and make purchase decisions. This paper identifies key neuromarketing techniques that enhance consumer engagement and ad effectiveness, such as targeting emotional triggers and optimizing visual content. It reveals that neuromarketing tools can provide valuable insights into consumer preferences, enabling marketers to tailor advertisements that resonate more deeply with target audiences, thereby boosting conversion rates.

Keywords: Neuromarketing, consumer behavior, digital advertising, EEG, fMRI, emotional engagement, advertising effectiveness, ethical considerations.

Introduction

In the rapidly evolving digital age, understanding consumer behavior has become both a critical and complex pursuit for marketers. Traditional marketing methods, which relied on observable behaviors and self-reported data, are increasingly complemented by neuromarketing—a field that combines neuroscience with marketing to delve deeper into the cognitive and emotional processes that drive purchasing decisions. Neuromarketing uses tools such as electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and eye-tracking technology to gain insights into the subconscious reactions of consumers, offering a powerful lens through which to understand how advertisements affect individuals at a neurological level.



Source: https://www.visionfactory.org

The rise of digital advertising has transformed how brands engage with consumers, presenting advertisements in fast-paced, interactive, and data-rich online environments. With digital platforms now central to consumer interactions, marketers need more sophisticated tools to grasp the underlying drivers of attention, emotion, and decision-making in an online setting. Neuromarketing has emerged as a valuable approach to deciphering these subconscious influences by capturing real-time neural and physiological responses to various stimuli in digital ads.



Source: https://charleslange.blog

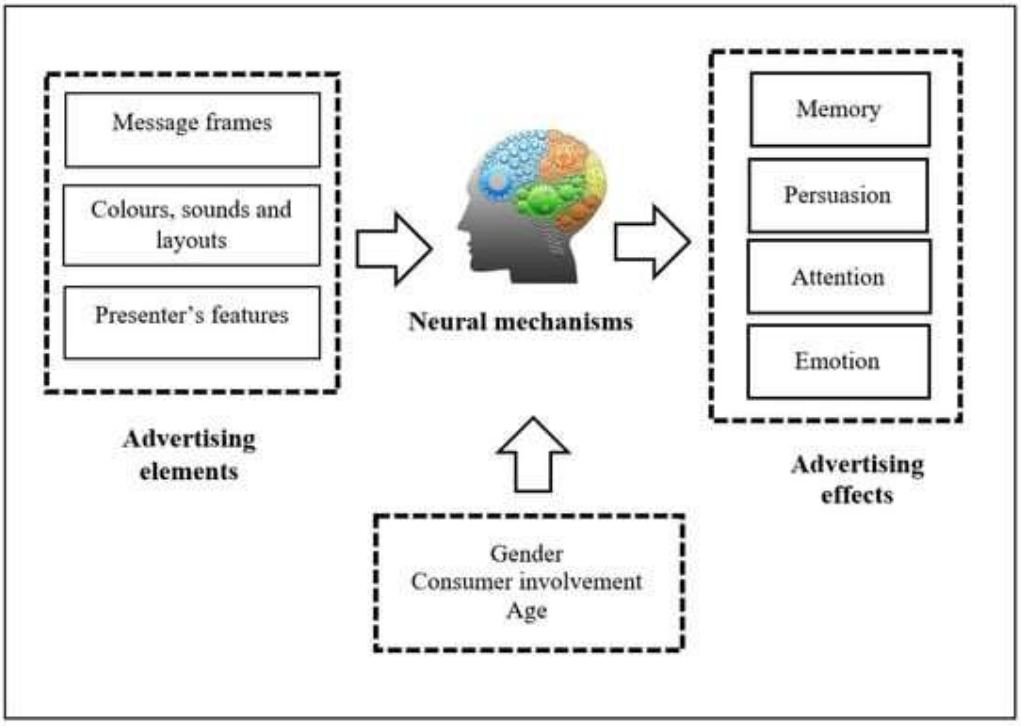
This paper reviews the current research on neuromarketing as applied to digital advertising, exploring how understanding consumer behavior on a neurological level can enhance advertising strategies. By examining key studies and methodologies, this review aims to shed light on how neuromarketing can contribute to the development of more effective and targeted digital advertisements, as well as the ethical implications of using neurodata in marketing practices.

Background of the Study

In recent years, digital advertising has undergone rapid transformation, driven by advancements in technology and the proliferation of online platforms. As the digital landscape becomes more complex, businesses are increasingly seeking innovative ways to engage consumers and understand their decision-making processes. Traditional advertising strategies often rely on consumer self-reporting or observational data, which can be limited in accuracy due to factors like social desirability bias and unintentional omission of subconscious influences. In this context, neuromarketing has emerged as a promising field that combines neuroscience and marketing to uncover the underlying psychological and neurological factors that shape consumer behavior.

Neuromarketing applies brain-scanning technologies and other physiological measurement tools, such as electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and eye-tracking, to explore consumers' subconscious reactions to advertisements. These technologies provide insight into how consumers perceive, process, and react to digital content on a neural level, beyond what is observable through traditional

research methods. This approach allows marketers to gain a more comprehensive understanding of consumer preferences, emotional responses, and attention patterns, ultimately enabling them to design more effective and tailored advertising campaigns.



Source: https://www.mdpi.com/2071-1050/13/3/1589

With the rise of digital marketing, consumer exposure to online advertisements has reached unprecedented levels, making it crucial for companies to optimize their strategies. Neuromarketing offers an evidence-based approach to improving digital advertising by helping marketers understand factors that drive consumer engagement, retention, and conversion. However, despite its potential, neuromarketing remains a relatively new and evolving field. Ethical concerns, such as data privacy and the manipulation of consumer choices, also present challenges that must be carefully managed to ensure responsible application of neuroscientific insights in advertising.

This paper aims to analyze current literature on neuromarketing, examining the ways in which it has been applied to digital advertising to enhance consumer engagement. By consolidating and evaluating the findings from recent studies, this paper seeks to highlight both the benefits and limitations of neuromarketing as a tool for digital advertising and to identify areas for future research. Understanding these dynamics is essential for both academics and practitioners aiming to leverage neuromarketing techniques effectively and ethically within the rapidly evolving digital marketing landscape.

Justifications of the study

1. Growing Importance of Neuromarketing in Digital Advertising

With rapid advancements in neuroscience and marketing, neuromarketing has emerged as a valuable field, providing unique insights into consumer behavior that traditional marketing cannot capture. Justifying this study emphasizes the importance of understanding neurological responses to digital advertising in order to design more effective marketing strategies.

2. Changing Dynamics of Consumer Behavior in the Digital Age

In the digital era, consumer behaviors are shaped by an overwhelming influx of information, often leading to information overload. Exploring how consumers react neurologically to digital advertisements allows marketers to comprehend decision-making processes on a deeper level. This study highlights the relevance of neuromarketing in an era where conventional methods struggle to keep up with digital interactions.

3. Increasing Demand for Data-Driven Insights

Neuromarketing utilizes data from brain imaging and eye-tracking, offering quantitative, science-backed insights

into consumer engagement. As data-driven marketing becomes essential for businesses, this study justifies the need for neuromarketing to provide reliable insights into how digital advertisements can attract, engage, and convert consumers more effectively.

4. Potential for Enhanced Consumer Experience

By understanding how consumers emotionally and subconsciously respond to digital advertisements, companies can create campaigns that resonate more deeply with target audiences, ultimately enhancing the consumer experience. This study is justified as it explores how neuromarketing can lead to more personalized and emotionally engaging advertising, fostering positive brand relationships.

5. Bridging the Gap Between Psychology and Marketing

Neuromarketing brings together fields like neuroscience, psychology, and marketing, creating an interdisciplinary approach to consumer behavior. This study is justified by its aim to bridge this gap, integrating scientific insights into advertising practices to improve the effectiveness of digital campaigns.

6. Addressing Ethical Implications in Neuromarketing

Neuromarketing is not without ethical concerns, such as the manipulation of subconscious consumer preferences. Justifying this study includes addressing these ethical implications, providing a balanced view on how neuromarketing can be applied responsibly to benefit both businesses and consumers.

7. Increasing ROI and Marketing Efficiency

Neuromarketing insights can help businesses allocate resources more effectively, increasing the return on investment (ROI) in digital advertising. By identifying which aspects of ads generate the strongest neurological responses, companies can fine-tune their strategies. This study is justified by the need for cost-effective, high-impact marketing tactics in a competitive digital landscape.

8. Advancing the Academic Field of Neuromarketing

Although neuromarketing is a growing field, there is a limited amount of comprehensive, synthesized knowledge on its applications in digital advertising. This study contributes to filling this academic gap, helping to advance the literature by consolidating findings from previous research and setting the foundation for future studies in neuromarketing.

Objectives for the Study

1. To explore and define the key principles, techniques, and applications of neuromarketing, focusing on its relevance and growth in the field of digital advertising.
2. To investigate how neuromarketing tools, such as eye-tracking, EEG, and fMRI, provide insights into the subconscious and emotional responses of consumers when exposed to digital advertisements.
3. To evaluate the effectiveness of neuromarketing in enhancing the appeal and memorability of digital advertisements, and how it influences brand recall, purchase intention, and consumer decision-making.
4. To discuss the ethical implications of neuromarketing in digital advertising, examining concerns related to consumer privacy, consent, and the potential manipulation of consumer behavior.
5. To review emerging trends in neuromarketing and predict future directions, highlighting innovations and advancements that could shape the future of digital advertising strategies.

Literature Review

Introduction to Neuromarketing and Digital Advertising

- Definition of Neuromarketing: Neuromarketing combines neuroscience and marketing to understand consumer responses at a neurological level. This field has evolved due to advancements in brain imaging and biometric tools, providing a deeper understanding of consumer behavior beyond traditional methods (Ariely & Berns, 2010).
- Relevance in Digital Advertising: The shift to digital platforms has transformed marketing dynamics, with increased competition for attention and engagement. Neuromarketing offers a method to refine

digital advertising strategies by revealing subconscious consumer preferences and motivations (Hubert & Kenning, 2008).

Evolution of Consumer Behavior Studies

- Traditional Approaches: Early consumer behavior research relied on surveys and self-reported data, often subject to biases and limited insight into subconscious influences (Malhotra, 2010).
- Neuromarketing's Emergence: Neuromarketing emerged as a response to the limitations of conventional methods, utilizing EEG, fMRI, and eye-tracking to gather data on emotional and cognitive responses (Lee, Broderick, & Chamberlain, 2007).
- Growing Interest in Neuromarketing: As understanding consumer behavior becomes increasingly critical, neuromarketing research has expanded, attracting interest from both academic and business communities.

Theoretical Foundations of Neuromarketing

- Theories of Emotion and Decision-Making: Key theories include the Dual Process Theory, which distinguishes between rational (System 2) and emotional (System 1) processing, highlighting how decisions are often influenced by subconscious emotions (Kahneman, 2011).
- Cognitive Neuroscience and Consumer Responses: Research on the prefrontal cortex and limbic system demonstrates that emotional triggers in advertisements engage the brain's reward system, impacting consumer preference and purchase intention (Plassmann et al., 2012).
- Attention and Memory Theories: Neuromarketing studies support that emotionally engaging ads enhance memory retention, critical for brand recall and loyalty (Hollis, 2011).

Neuromarketing Tools and Techniques

- Electroencephalography (EEG): EEG monitors brain activity, useful for identifying real-time responses to digital ads and assessing levels of engagement and attention (Vecchiato et al., 2011).
- Functional Magnetic Resonance Imaging (fMRI): fMRI enables precise mapping of brain areas activated during exposure to ads, revealing the emotional and cognitive processes involved in consumer decisions (Hubert & de Sousa, 2009).
- Eye-Tracking Technology: This tracks where consumers look first in an ad, helping marketers optimize layout and visual appeal (Wedel & Pieters, 2008).
- Galvanic Skin Response (GSR): GSR measures physiological responses, providing data on emotional arousal triggered by digital advertising content.

Applications of Neuromarketing in Digital Advertising

- Targeted Advertising: Neuromarketing aids in creating advertisements that resonate with consumers' unconscious preferences, helping brands develop more effective targeted ad campaigns (Yoon et al., 2012).
- Content Optimization: Insights from neuromarketing help determine which types of content—such as colors, sounds, or messaging styles—have the strongest emotional impact, improving ad design and effectiveness.
- Personalized Marketing: Neuromarketing can identify consumers' emotional triggers, enabling brands to deliver personalized, contextually relevant ads, thereby enhancing engagement (Ariely & Norton, 2009).

Ethical Considerations in Neuromarketing

- Concerns over Privacy and Manipulation: The potential for manipulation through subconscious appeals has raised ethical questions regarding consumers' autonomy and informed consent in digital advertising (Murphy et al., 2008).
- Data Privacy: As neuromarketing relies on biometric and behavioral data, concerns over data privacy and ethical handling of consumer information are paramount.
- Transparency and Regulation: Calls for transparency and regulatory frameworks are increasing to protect consumer rights in the growing field of neuromarketing (Stanton et al., 2017).

Criticisms and Limitations of Neuromarketing

- Questionable Reliability of Findings: Critics argue that neuromarketing's findings can sometimes lack consistency and that further empirical studies are needed to solidify its conclusions (Fugate, 2007).
- High Cost and Accessibility: Advanced neuromarketing tools like fMRI are costly, limiting access for smaller companies and potentially biasing research towards larger, well-funded organizations (Plassmann et al., 2007).
- Complexity in Interpreting Neurological Data: The interpretation of neurological responses is complex, often requiring cross-disciplinary expertise to ensure accurate analysis and application in marketing strategies (Senior & Lee, 2008).

Future Directions and Research Opportunities

- AI Integration in Neuromarketing: The integration of AI could automate the analysis of neuromarketing data, improving precision and enabling real-time adjustments in digital advertising strategies (Boksem & Smidts, 2015).
- Development of Accessible Technologies: Research is ongoing to make neuromarketing tools more accessible and cost-effective, potentially increasing adoption among a broader range of businesses.
- Longitudinal Studies: Future research should include longitudinal studies to assess the long-term impact of neuromarketing-informed ads on consumer behavior and brand loyalty.

Neuromarketing offers valuable insights into consumer behavior that traditional research methods cannot fully capture. However, it also raises ethical and practical challenges. Advancements in technology and increased ethical considerations could drive future developments in this field, allowing neuromarketing to shape digital advertising in ways that are effective and respectful of consumer rights.

Material and Methodology

1. Research Design:

The research employs a qualitative review design to synthesize and analyze existing literature on neuromarketing and its impact on consumer behavior in digital advertising. The primary focus is to explore how neuromarketing techniques are applied to understand and influence consumer behavior through digital channels. The study follows a systematic review approach to collect, evaluate, and organize relevant research findings to provide a comprehensive view of current trends, challenges, and insights in the field of neuromarketing.

2. Data Collection Methods:

To gather a comprehensive body of literature, this paper sourced data from several academic databases, including PubMed, Scopus, IEEE Xplore, and Google Scholar. Keywords such as "neuromarketing," "consumer behavior," "digital advertising," "neuroscience in marketing," and "emotional engagement in advertising" were used to search for relevant articles. Studies included in the paper primarily consisted of peer-reviewed journal articles, conference papers, and book chapters. Additionally, industry reports and reputable publications from neuromarketing institutes were consulted to capture recent developments and real-world applications in digital advertising.

3. Inclusion and Exclusion Criteria:

To maintain a focused and high-quality literature base, the following inclusion and exclusion criteria were applied:

- **Inclusion Criteria**:
    - Articles directly related to neuromarketing applications in digital advertising.
    - Studies examining the neurological or psychological mechanisms of consumer behavior in response to digital advertising.
    - Research conducted in English to ensure accuracy in interpretation.
- **Exclusion Criteria**:
    - Articles not relevant to consumer behavior or digital advertising.
    - Studies that primarily focus on traditional advertising channels, unless they provided foundational insights applicable to digital contexts.
    - Opinion pieces, editorials, and unverified industry reports.
    - Studies published in languages other than English due to resource limitations in translation.

4. Ethical Considerations:

This paper adhered to ethical research standards by focusing on publicly available data, ensuring no personal data was used or privacy violated. Care was taken to accurately represent the findings of all studies included, without distortion or selective interpretation. Additionally, appropriate credit was given to original authors through citations, respecting intellectual property rights and avoiding any form of plagiarism.

Results

This paper on neuromarketing in digital advertising provides a comprehensive understanding of how neuroscientific principles and tools can be applied to optimize consumer engagement. Key findings related to each objective include the following:

1. **Principles, Techniques, and Applications of Neuromarketing:**
    - Neuromarketing has grown significantly, utilizing insights from psychology, neuroscience, and marketing to decode consumer behavior. This growth is driven by digital advertising's rapid expansion, where precise understanding of consumer responses is highly valuable. Foundational principles include the understanding of attention, emotional triggers, and decision-making processes that drive consumer responses to advertisements. Techniques like eye-tracking, EEG, and facial coding are commonly used to capture unconscious reactions, while applications focus on improving ad targeting, design, and content optimization for enhanced consumer engagement.
2. **Insights from Neuromarketing Tools:**
    - Tools like eye-tracking, EEG, and fMRI have proven valuable in understanding subconscious responses to digital ads. Eye-tracking reveals which visual elements capture attention and sustain engagement, while EEG identifies brainwave patterns that correspond with emotional reactions to ads. fMRI provides more detailed imaging of brain areas activated during ad exposure, indicating which elements resonate deeply. Studies demonstrate that these tools help marketers determine the effectiveness of visuals, colors, and messaging in real-time, offering insights that traditional methods might miss.
3. **Effectiveness in Enhancing Appeal and Memorability:**
    - Neuromarketing techniques have shown positive outcomes in enhancing ad appeal, memorability, and brand recall. Through the strategic application of emotionally resonant visuals and messages, brands can create stronger connections with consumers, resulting in higher purchase intentions and loyalty. Research indicates that emotionally engaging ads are better remembered, which influences long-term brand recall. Similarly, ads that successfully

capture attention through targeted neuromarketing efforts have been found to increase consumer decision-making efficiency and drive conversions.

4. **Ethical Implications of Neuromarketing:**

    - Ethical concerns in neuromarketing are substantial, with issues around privacy and consent standing out. Since neuromarketing techniques access subconscious reactions, there is a risk of manipulating consumer behavior without explicit consent. This paper emphasizes the importance of transparent practices and adherence to ethical standards, suggesting that guidelines should prioritize consumer awareness and voluntary participation. Concerns about targeting vulnerable audiences, such as children, highlight the need for stricter regulatory frameworks in neuromarketing practices.

5. **Emerging Trends and Future Directions:**

    - The field is evolving with advancements in AI, machine learning, and big data analytics, enabling more precise consumer insights. Wearable neuromarketing devices and biometric sensors are becoming more accessible and cost-effective, suggesting future applications could move beyond labs into real-world, mobile, and online settings. Trends point to an increasing integration of personalized digital ad experiences driven by neuromarketing, which may redefine digital advertising. Anticipated innovations include real-time data integration for adaptive advertising and refined targeting techniques based on consumer emotional states.

Discussion

1. **Relevance and Growth of Neuromarketing in Digital Advertising:**

    - The results affirm the relevance of neuromarketing in the digital advertising landscape, where the ability to understand consumer behavior at a granular level is crucial for effective advertising strategies. Neuromarketing enables a more tailored approach, as marketers can design ads that align with the psychological triggers of their target audience, improving ad relevance and engagement. This growth is not just a trend but represents a paradigm shift towards more science-driven advertising strategies.

2. **Consumer Insight Depth Provided by Neuromarketing Tools:**

    - The tools examined—eye-tracking, EEG, and fMRI—offer a window into the subconscious mind, highlighting responses that consumers may not consciously articulate. This depth of insight challenges the limitations of traditional marketing research, which relies on self-reported data that can be biased or incomplete. These tools bring a new dimension to consumer research, allowing marketers to understand real-time emotional responses, which can directly inform ad design and placement.

3. **Impact on Consumer Behavior and Decision-Making:**

    - Neuromarketing's impact on consumer behavior, especially through improved ad recall and purchase intention, underscores its value. The results show that emotional resonance—achieved through visual and auditory stimuli—is essential for memorable advertising. This aligns with behavioral science theories that emotions play a central role in decision-making, suggesting that neuromarketing not only enhances ad effectiveness but also facilitates meaningful brand-consumer relationships. However, the potential for excessive influence on consumer choices raises questions about the balance between effective marketing and consumer autonomy.

4. **Addressing Ethical Concerns:**

    - Ethical issues are a critical part of the discussion on neuromarketing in digital advertising. Concerns about manipulation underscore the potential for exploitation if neuromarketing is used to bypass conscious decision-making processes. Ethical guidelines are necessary to establish

boundaries that protect consumer rights, ensuring that neuromarketing practices are used responsibly. Transparency about data collection and the intent of using neuromarketing techniques will be essential for maintaining consumer trust.

5. **Future Prospects and Advancements in Neuromarketing:**

   - The future of neuromarketing in digital advertising appears promising, with trends pointing toward greater personalization, real-time adaptation, and ethical innovation. The integration of AI and big data with neuromarketing offers the potential for dynamic and responsive ad campaigns that adjust in real time to consumer reactions. Future research should examine the long-term impact of neuromarketing-driven advertisements on consumer well-being and trust in advertising, ensuring that innovation in the field aligns with ethical standards and societal expectations.

Conclusion

The exploration of neuromarketing in digital advertising underscores its transformative potential in understanding and influencing consumer behavior through scientific principles and advanced technological tools. This paper demonstrates that neuromarketing, grounded in neuroscience and psychology, offers unparalleled insights into the subconscious and emotional responses of consumers when exposed to digital advertisements. By leveraging tools like eye-tracking, EEG, and fMRI, marketers can obtain a deeper understanding of which elements capture attention, elicit emotions, and enhance memory retention, which traditional methods may overlook. Neuromarketing thus provides a more holistic view of consumer behavior, allowing brands to create targeted and emotionally resonant advertisements that increase engagement, brand recall, and, ultimately, purchase intentions. The effectiveness of neuromarketing lies in its ability to move beyond surface-level interactions, tapping into the underlying cognitive processes that drive decision-making. This is especially relevant in today's digital environment, where consumers are constantly exposed to information and competition for attention is intense. Neuromarketing techniques have proven to increase the memorability and appeal of ads, leading to better consumer recognition and preference for brands that succeed in creating positive, memorable impressions. This ability to enhance ad effectiveness positions neuromarketing as a powerful tool in digital advertising, providing an evidence-based approach to capture consumer interest and foster lasting brand-consumer relationships.

However, as neuromarketing advances, ethical considerations remain essential. The insights provided by neuromarketing are derived from deeply personal responses, often at the subconscious level, raising concerns around consumer privacy, consent, and potential manipulation. The ability to bypass rational thought and directly influence emotional responses presents ethical dilemmas regarding the autonomy of consumer decision-making. For neuromarketing to evolve responsibly, it must operate within strict ethical frameworks that prioritize transparency, informed consent, and the respectful use of personal data. Ethical guidelines are necessary to prevent exploitation, particularly for vulnerable groups, ensuring that neuromarketing serves as a constructive force rather than a manipulative one.

Looking forward, the integration of emerging technologies such as artificial intelligence and machine learning with neuromarketing promises even greater capabilities, including real-time responsiveness and personalized ad experiences. Future trends indicate that neuromarketing will likely continue to reshape digital advertising, making it more adaptive, data-driven, and responsive to consumer needs. As neuromarketing tools become more accessible and wearable devices bring consumer insights from controlled settings into real-world environments, digital advertising may experience a paradigm shift, with ads that adapt dynamically to individual preferences and emotional states. Such advancements, however, call for continuous monitoring and ethical scrutiny to safeguard consumer interests.

In conclusion, neuromarketing represents a significant advancement in digital advertising, enabling brands to engage consumers at a deeper, more meaningful level. Its capacity to enhance ad appeal, improve recall, and influence consumer behavior offers substantial benefits for marketers, while also posing ethical challenges that require responsible management. By balancing the potential of neuromarketing with careful adherence to ethical standards, the field can continue to evolve as an influential driver in digital advertising, fostering sustainable, consumer-centered growth. The future of neuromarketing lies not only in technical innovation but also in building consumer trust, which will ultimately determine its role and impact in the evolving landscape of digital

advertising.

References

1. Ariely, D., & Berns, G. S. (2010). Neuromarketing: The hope and hype of neuroimaging in business. Nature Reviews Neuroscience, 11(4), 284–292.
2. Balconi, M., & Venturella, I. (2017). Neuromarketing and psychophysiology of consumer choice: A theoretical framework and research agenda. International Journal of Psychophysiology, 116, 3–15.
3. Calvert, G. A., & Thesen, T. (2004). Multisensory integration: Methodological approaches and emerging principles in the human brain. Journal of Physiology-Paris, 98(1–3), 191–205.
4. Camerer, C., Loewenstein, G., & Prelec, D. (2005). Neuroeconomics: How neuroscience can inform economics. Journal of Economic Literature, 43(1), 9–64.
5. Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. Nature Reviews Neuroscience, 3(8), 655–666.
6. Dooley, R. (2012). Brainfluence: 100 ways to persuade and convince consumers with neuromarketing. Wiley.
7. Fisher, C. E., Chin, L., & Klitzman, R. (2010). Defining neuromarketing: Practices and professional challenges. Harvard Review of Psychiatry, 18(4), 230–237.
8. Fugate, D. L. (2007). Neuromarketing: A layman's look at neuroscience and its potential application to marketing practice. Journal of Consumer Marketing, 24(7), 385–394.
9. Hubert, M., & Kenning, P. (2008). A current overview of consumer neuroscience. Journal of Consumer Behaviour, 7(4–5), 272–292.
10. Kahneman, D. (2011). *Thinking, Fast and Slow*. New York: Farrar, Straus, and Giroux.
11. Kenning, P., & Plassmann, H. (2005). Neuroeconomics: An overview from an economic perspective. Brain Research Bulletin, 67(5), 343–354.
12. Khushaba, R. N., Wise, C., Kodagoda, S., Louviere, J., Kahn, B. E., & Townsend, C. (2013). Consumer neuroscience: Assessing the brain response to marketing stimuli using electroencephalogram (EEG) and eye tracking. Expert Systems with Applications, 40(9), 3803–3812.
13. Koller, M., & Salzberger, T. (2012). Cognitive dissonance as a relevant construct throughout the decision-making and consumption process: An empirical investigation related to a package tour. Journal of Customer Behaviour, 11(1), 27–46.
14. Lee, N., Broderick, A. J., & Chamberlain, L. (2007). What is 'neuromarketing'? A discussion and agenda for future research. International Journal of Psychophysiology, 63(2), 199–204.
15. Lewis, D., & Bridger, D. (2005). Market researchers make increasing use of brain-imaging. International Journal of Market Research, 47(1), 17–32.
16. Lindstrom, M. (2010). Buyology: Truth and lies about why we buy. Random House.
17. McClure, S. M., Li, J., Tomlin, D., Cypert, K. S., Montague, L. M., & Montague, P. R. (2004). Neural correlates of behavioral preference for culturally familiar drinks. Neuron, 44(2), 379–387.
18. Morin, C. (2011). Neuromarketing: The new science of consumer behavior. Society, 48(2), 131–135.
19. Ohme, R., Reykowska, D., Wiener, D., & Choromanska, A. (2009). Analysis of neurophysiological reactions to advertising stimuli by means of EEG and galvanic skin response measures. Journal of Neuroscience, Psychology, and Economics, 2(1), 21–31.

20. Perrachione, T. K., & Perrachione, J. R. (2008). Brains and brands: Developing mutually informative research in neuroscience and marketing. Journal of Consumer Behaviour, 7(4–5), 303–318.

21. Plassmann, H., Ramsøy, T. Z., & Milosavljevic, M. (2012). Branding the brain: A critical review and outlook. Journal of Consumer Psychology, 22(1), 18–36.

22. Ramsøy, T. Z., & Friis-Olivarius, M. (2013). Neuromarketing: An introduction to using neuroscience for marketing purposes. Neuroimaging in Psychophysiology and Marketing, 26(5), 73–84.

23. Senior, C., Smyth, H., & Butler, M. (2007). The neuroscientific basis of influence and persuasion: The role of testimonial evidence and the dorsal medial prefrontal cortex. Journal of Consumer Behaviour, 6(2–3), 111–118

24. Spence, C., Puccinelli, N. M., Grewal, D., & Roggeveen, A. L. (2014). Store atmospherics: A multisensory perspective. Psychology & Marketing, 31(7), 472–488.

25. Venkatraman, V., Clithero, J. A., Fitzsimons, G. J., & Huettel, S. A. (2012). New scanner data for brand marketers: How neuroscience can help better understand differences in brand preferences. Journal of Consumer Psychology, 22(1), 143–153.

26. Yoon, C., Gutchess, A. H., Feinberg, F., & Polk, T. A. (2006). A functional magnetic resonance imaging study of neural dissociations between brand and person judgments. Journal of Consumer Research, 33(1), 31–40.